

**Comparison of Functional outcome Closed Reduction and
Percutaneous Pinning versus complete cast for management of
distal radius fracture with disrupted radioulnar joint**



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INTRODUCTION

Distal radius fractures are among the most prevalent orthopedic emergencies, significantly impacting patient quality of life.¹ Distal radius breaks speak to the 10%-12% of all cracks of the body and as much as 72% of all lower arm cracks. Distal radioulnar joint shakiness is a significant intricacy of distal span break and is significant reason for ulnar sided wrist torment in distal end sweep cracks.² This type of fracture significantly impacts patient quality of life, necessitating effective management strategies to restore function and anatomy.¹

Decreased range of motion of the elbow and forearm and decreased grip strength are potential findings following a childhood upper extremity fracture. Clinical follow-up is essential because spontaneous improvement is seen several months after the injury. Freehand measuring with a goniometer and hydraulic dynamometer is used to evaluate clinical result. The new methods are justified in avoiding human typewriting errors, thus improving patient safety.³ The distal radius, with its critical role in wrist and hand movements, requires precise therapeutic interventions to ensure the best possible outcomes.⁴

A key method of surgical fixation is percutaneous pinning, involving the insertion of wires through the skin to stabilise the fracture. In most pinning methods, pins or wires are used to fix the fragments together.⁵ Closed reduction and cast treatment of distal radius fractures renders satisfactory results in fractures that are reducible and stable and do not re-displace in plaster in the first 2 weeks following reduction.^{6, 7}

Karadeniz et al., conducted a trial and observed that flexion was more with cast 82.58 ± 30.50 as compared to CRPP (81.05 ± 26.42), as well as Extension (95.27 ± 17.04) vs. (93.43 ± 15.02). But the difference in both groups was insignificant ($p > 0.05$). The mean grip strength was less

with cast (94.06 ± 26.17) as compared to CRPP (95.49 ± 31.41), although the difference was insignificant ($p=0.681$).⁸ Rodríguez-Merchán conducted a trial and observed that functional results in the pinning group were better (excellent, 60%; good, 30%; fair, 10%) than in the plaster group (excellent, 15%; good, 40%; fair, 25%; poor, 20%). Anatomic results also were better in the pinning group ($p<0.05$).⁹

Rationale of this study is to compare the functional outcome CRPP versus complete cast for management of distal radius fracture with disrupted radioulnar joint. Literature showed that CRPP is a reliable method and has more excellent functional outcome as compared to plaster cast. But in routine, for such cases, plaster cast is applied and patients is conservatively managed. This may be due to lack of evidence in local population. Therefore, there is a need to find the evidence for local population. So through this study to will get the data and beneficial aspects of CRPP and replace plaster cast in distal radius fracture in local population. This will help to improve our practice and outcome of treatment for distal radius fracture with disrupted radioulnar joint.

OBJECTIVE:

To compare the functional outcome closed reduction and percutaneous pinning versus complete cast for management of distal radius fracture with disrupted radioulnar joint

OPERATIONAL DEFINITIONS

Disrupted radioulnar joint: It is defined as an abnormal alignment of radius and ulna bones $>5\text{mm}$ in the wrist due to fracture of distal end of radius, assessed by X-ray.

Outcome: It will be assessed in terms of following:

- **Mean range of motion:** It will be assessed as extension and flexion of wrist joint (radioulnar joint) after 4 and 6 weeks of treatment
- **Mean Grip strength:** It will be assessed by using hand grip dynamometer after 4 and 68 weeks of treatment (annexure I)
- **Functional outcome:** It will be assessed by using grip strength assessment tool in terms of excellent, good, fair and poor after 2, 4 and 6 weeks of treatment

HYPOTHESIS:

There is a difference in the functional outcome closed reduction and percutaneous pinning versus complete cast for management of distal radius fracture with disrupted radioulnar joint

MATERIAL AND METHODS

Study Design: Randomized controlled trial

Setting: Department of Orthopedic Surgery, Aziz Bhatti Shaheed Teaching Hospital, Gujrat

Duration: 6 months after the approval of synopsis from CPSP.

Sample Size: By using WHO calculator, sample size of 60 cases; 30 in each group is calculated with 5% significance level, 90% power of study and percentage of excellent grip functional outcome i.e. 60% with CRPP and 15% with complete cast.⁹

Sampling technique: Non-probability, consecutive sampling

Sample Selection

Inclusion Criteria: Patients of age 16-75 years, both genders, presenting with distal radius fracture with disrupted radioulnar joint (as per operational definition).

Exclusion Criteria: Patients with coagulation disorder (PT>15 sec), connective tissue disorder, multiple fractures at same site, recurrent fracture, previous failed surgery, open fracture (on clinical examination)

Data Collection Procedure: After approval from ethical review board, ## patients fulfilling the selection criteria will be included in this study from emergency department. Informed consent will be taken. Demographics (name, age, gender, anatomical side, h/o diabetes (BSR>200 mg/dl), hypertension (BP $\geq$ 140/90 mmHg), mode of injury, duration of fracture and pain score at presentation) will also be noted. Then patients will be randomly divided in two groups by using lottery method. In group A, patients will be treated with closed reduction and percutaneous pinning. Pins will be removed after 6 weeks. In group B, patients will be treated with complete plaster cast. All the procedures will be done by consultant surgeon with assistance of researcher. Patients will be followed-up in OPD for 6 weeks. On each visit, patients will be examined clinically and radiologically. On each visit, range of motion and grip strength by using hand grip dynamometer will be measured using a scoring system reported by Home et al., and functional outcome will be noted as excellent, good, fair or poor (as per operational definition). All the data will be collected in proforma (attached).

Statistical analysis: All the data will be entered and analyzed through SPSS version 26.0. Normality of data will be checked by using Shapiro-Wilk test. The quantitative variables like age, duration of fracture, pain score at presentation, post-treatment functional outcome score will be presented as mean & SD. The qualitative variable like gender, anatomical side, history of diabetes, hypertension, mode of injury, and excellent and good functional outcome will be presented as frequency and percentage.

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PROFORMA

Comparison of Functional outcome Closed Reduction and Percutaneous Pinning versus complete cast for management of distal radius fracture with disrupted radioulnar joint

Case No. _____ Hosp. Reg. No. _____ Date _____

Name _____ Age _____

Gender M ☐ F ☐ Duration of injury: _____

Anatomical Side: Left ☐ Right ☐

Mode of injury: _____

History: Diabetes ☐ Hypertension ☐

Pain score at presentation: _____ (VAS)

Group: CRPP ☐ Complete cast ☐

FOLLOW UP:

	After 2 weeks	After 4 weeks	After 6 weeks
Range of motion			
Grip strength			

ANNEXURE I

Grip strength assessment

Female (Age)	Needs Improvement	Fair	Good	Very Good	Excellent
15-19	< 54	54-58	59-63	64-70	> 70
20-29	< 55	55-60	61-64	65-70	> 70
30-39	< 56	56-60	61-65	66-72	> 72
40-49	< 55	55-58	59-64	65-72	> 72
50-59	< 51	51-54	55-58	59-64	> 64
60-69	< 48	48-50	51-53	54-59	> 59

Male (Age)	Needs Improvement	Poor	Fair	Good	Excellent
15-19	< 84	84-94	95-102	103-112	> 112
20-29	< 97	97-105	106-112	113-123	> 123
30-39	< 97	97-104	105-112	113-122	> 122
40-49	< 94	94-101	102-109	110-118	> 118
50-59	< 87	87-95	96-101	102-109	> 109
60-69	< 79	79-85	86-92	93-101	> 101